Student Number

Sri Lanka Institute of Information Technology

B.Sc. (Hons)
in
Computer Science

Final Examination Model Paper

Design and Analysis of Algorithms (SE2022)

Duration: 2 Hour

Instruction to Candidates:

QUESTION ONE

(1) List two properties of good algorithm.

(2) What is the main purpose of analysis of algorithms?

(3) Analyze the running time of the following program fragment assuming a RAM model of computation.

```
j ← 0
for i ← 0 to n
    while j ≤ 5
        j ←j +1
```

(4) What are the best case and worst case running time (in Big O notation) of the insertion sorting algorithm? When do you get such situations in the insertion sorting algorithm?

(5) Illustrate the operations of the *Insertion sort algorithm* for the array with the given set of elements. (For the illustration process assign the values only once to the given algorithm codes and then use diagrammatic way to reach the answer.)

A

1	2	3	4
10	0	8	4

```
INSERTION-SORT(A)
1 for j ←2 to length[A]
2    do key ← A[j]
3       ▷ Insert A[j] into the sorted sequence A[1..j-1]
4       i ← j - 1
5       While i > 0 and A[i] > key
6             do  A[i+1] ← A[i]
7                 i ←i-1
8       A[i+1] ← key
```

(6) Find the *Big oh* value for the following functions. Justify your answer. (02 marks)

a) $T(n) = \log n + n + 100$

b) $T(n) = 200n^{20} + 100n^{10} + 2^n + 3^n + n + 1$

(7) The power function can be defined as $pow(x, n) = x^n$. This can be evaluated using the multiplication as $x^n = x \times x^{n-1}$ where x is any real number and n is a non-negative integer. [Hint: $pow(x, n-1) = x^{n-1}$] (06 marks)

a) Write a recursive relation for $pow(x, n)$ where x is any real number and n is a non-negative integer. Clearly define the initial condition(s).
b) Write a recursive algorithm in pseudo code for the above recursive relation.
c) Write a recurrence equation that describe the running time T (n) for the above part b) recursive algorithm.

QUESTION TWO

(1) Given below is an algorithm for *QUICKSORT* (*A, p, r*)

```
Procedure QUICKSORT (A, p, r)
1  if p < r
2    then q ← PARTITION(A, p, r)
3         QUICKSORT (A, p, q-1)
4         QUICKSORT (A, q+1, r)

Procedure PARTITION(A, p, r)
1  x ← A[r]
2  i ← p - 1
3    for j ← p to r - 1
4       do if A[j] ≤ x
5             then i ← i + 1
6                  exchange A[i] ↔ A[j]
7    exchange A[i + 1] ↔ A[r]
8  return i + 1
```

a) Illustrate the operations of the *QUICKSORT (A, p, r)* for the array with the given set of elements. (For the illustration process assign the values only once to the given algorithm codes and then use diagrammatic way to reach the answer.) (03 marks)

1	2	3	4	5	6
28	15	1	30	0	8

b) Depending on the way array is partitioned by the quick sort algorithm, best case or worst case can occur. What kinds of partitioning generate the best or worst case? (02 marks)

c) Write recurrence equations for best case in the above part b).

d) Use the repeated substitution method to solve the best case recurrence equation.

marks)

Qestion three (14 marks)

1) Briefly describe the following terms as related to tree terminology.
   a) Leaf node
   b) Binary Tree

(2 marks)

2) Prove that the relationship between number of nodes ***n*** and the height ***h*** of a Full Binary Tree is given by $n+1=2^{h+1}$.

(2 marks)

3) Illustrate the operations of the Heap sort for the array **A** of elements given below. (For the purpose of illustration, assign the values only once to the given algorithms at the end of the paper and use diagrammatic way to reach the answer.)

	1	2	3	4	5	6	7	8
A	45	25	15	85	75	65	35	55

(6 marks)

4) Is the sequence represent <23, 17, 14, 6, 13, 10, 1, 5, 7, 12> a max heap? Justify your answer.

(2 marks)

5) CPU scheduling can be done based on priority assigned by the operating system (OS). Therefore the OS has to use a priority queue for the CPU scheduling. Assumed that the priority is represented by a number for each process and high number means high priority.

6) What type of heap should be used for the implementation of the above priority queue?

Question four (11 Marks)

1) Draw the Directed graph if V={1,2,3,4,5}and E={(1,2),(2,3),(3,4),(3,5),(5,4),(3,2)}.

(1 mark)

2) There are two methods to represent the graph in a computer system. Briefly explain the two methods. Which representation is more suitable for Sparse Graph? Justify your answer.

Question One (14 marks)

7) Briefly describe the following terms as related to tree terminology.
   a) Leaf node
   b) Binary Tree

(2 marks)

8) Prove that the relationship between number of nodes ***n*** and the height ***h*** of a Full Binary Tree is given by $n+1=2^{h+1}$.

(2 marks)

9) Illustrate the operations of the Heap sort for the array **A** of elements given below. (For the purpose of illustration, assign the values only once to the given algorithms at the end of the paper and use diagrammatic way to reach the answer.)

	1	2	3	4	5	6	7	8
A	45	25	15	85	75	65	35	55

(6 marks)

10) Is the sequence represent <23, 17, 14, 6, 13, 10, 1, 5, 7, 12> a max heap? Justify your answer.

(2 marks)

11) CPU scheduling can be done based on priority assigned by the operating system (OS). Therefore the OS has to use a priority queue for the CPU scheduling. Assumed that the priority is represented by a number for each process and high number means high priority.

12) What type of heap should be used for the implementation of the above priority queue?

Question Two (11 Marks)

4) Draw the Directed graph if V={1,2,3,4,5}and E={(1,2),(2,3),(3,4),(3,5),(5,4),(3,2)}.

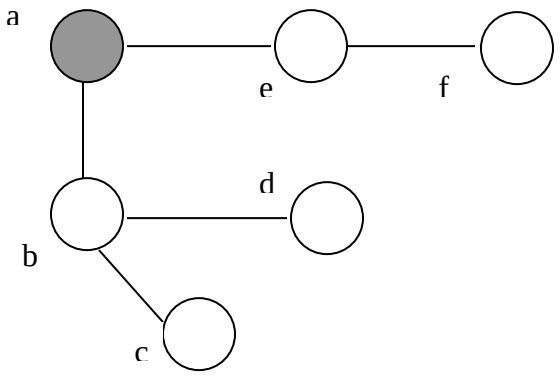
(1 mark)

5) There are two methods to represent the graph in a computer system. Briefly explain the two methods. Which representation is more suitable for Sparse Graph? Justify your answer.

(3 marks)

6) Illustrate the operation of the DFS (Depth First Search) on the graph given below taking the starting vertex as vertex ***a***. (For the purpose of illustration, assign the values only once to the given algorithms at the end of the paper and use diagrammatic way to reach the answer.)

(6 marks)



(12 Marks)

3) What is meant by fixed length coding in data representation? Give one example for fixed length coding.

(2 marks)

4) A file contains only digits in the following frequency:

Digits	0	1	2	3	4	5
Frequencies	121	50	203	401	8	195

e) Construct the codeword for the above data using Huffman Algorithm given at the end of the paper.

(4 marks)

f) How many bits are needed to represent the digits in the file?

(1 mark)

g) Construct the codeword for the above data using Shannon Fano Algorithm given at the end of the paper (page 16).

(4 marks)

h) Which codeword is optimal out of the above constructed codeword in (a) and (c)?

(1 mark)

Question Five (20 Marks)

4) What is meant by optimal substructure in Dynamic Programming?

(1 mark)

5) Construct a table for the following 0/1 knapsack problem using Dynamic Programming with the recursive solution given at the end of the paper (page 16).

$n = 4,\ c = 10,\ w = [\ 3, 4, 6\ ,\ 3\ ]$ and $p = [\ 6, 12, 8, 2\ ]$

(8 marks)

c) Find $p(3,9)$ and $p(1,10)$ using the recursive equations given at the end of the paper (page 16).

(2 marks)

d) Briefly explain how you achieved the final solution.

(2 marks)

6) Given a chain $(A_1, A_2, ..., A_{n-1}, A_n)$ of n matrices, where for i = 1,2, …, n matrix A_i has dimension $p_{i-1} \times p_i$. Assume that $m[i,j]$ is the minimum number of scalar multiplications needed to compute the matrix $A_{i...j} = A_i \times A_{i+1} \times ..... \times A_{j-1} \times A_j$ and it is defined below

$$m[i, j] = \begin{cases} 0 & \text{if } i = j \\ \min\limits_{i \le k < j} \{m[i,k] + m[k+1, j]\} + p_{i-1} p_k p_j & \text{otherwise} \end{cases}$$

Consider the following set of metrics A_1, A_2, A_3 and A_4 with their dimensions of $2 \times 5,\ 5 \times 3,\ 3 \times 10$ and 10×4 respectively.

a) Draw the *m* and *s* table to find the optimal parenthesizing of the matrices for the above sequence of matrices using the Dynamic Programming algorithm.

b) Hence find the optimal parenthesizing and optimal number of scalar multiplications of the above matrices

(7 marks)

End of Question Paper

Algorithms

```
Procedure HEAPSORT(A)
1.BUILD_HEAP[A]
2.for  i ← length[A] down to 2
3. Exchange A[1] ↔ A[i]
4. heap_size[A] ← heap_size[A]-1;
5. HEAPIFY(A,1)
```

```
Procedure BUILD_HEAP (A)
1.heap_size[A] ← length[A]
2.  for i ← ⌊length[A]/2⌋ downto 1
3.          HEAPIFY(A,i)
```

```
Procedure HEAPIFY (A,i)
1.  l ← LEFT_CHILD (i);

2.  r ← RIGHT_CHILD (i);

3.  if l ≤ heap_size[A] and A[ l ] > A[ i ]
4.          then largest ← l;
5.          else largest ← i;
6.  if r ≤ heap_size[A] and A[r] > A[largest]
7.          then largest ← r;
8.  if largest ≠ i
9.          then exchange A[i] ↔ A[largest]
10.             HEAPIFY (A, largest)
```

```
Procedure DFS(G)
1   for each vertex u ∈V [G]
2               do color[u] ← WHITE
3                       π[u] ← NIL
4   time ← 0
5   for each vertex u ∈V [G]
6               do if color[u] = WHITE
7                       then DFS-VISIT(u)

Procedure DFS-VISIT(u)
1   color[u] ← GRAY     .
2   time ← time +1
3   d[u]  ← time
4   for each v ∈Adj[u]    ▷Explore edge (u, v).
5               do if color[v] = WHITE
6                       then π[v] ← u
7                               DFS-VISIT(v)
8   color[u] ← BLACK     ▷Blacken u; it is finished.
9   time ← time +1
10  f [u] ← time
```

```
Huffman (C)
1       n ← |C|
2       Q ← C
3       for i ← 1 to n-1
4               do allocate a new node z
5               left[z] ← x ← EXTRACT-MIN(Q)
6               right[z] ← y ← EXTRACT-MIN(Q)
7               f[z] ← f[x] + f[y]
8                       INSERT(Q,z)
9       return  EXTRACT-MIN(Q)
```

In the 0/1 knapsack problem we define P(i,k) to be the maximum profit possible using items i…n and remaining capacity k. We denote the weight and profit of the ith item by w_i and p_i .

$$P(i,k)=\begin{cases}0 & i=n \;\&\; w_n > k\\ p_n & i=n \;\&\; w_n \le k\\ P(i+1,k) & i<n \;\&\; w_i > k\\ \max(P(i+1,k),\ p_i + P(i+1,k-w_i)) & i<n \;\&\; w_i \le k\end{cases}$$

Rough Sheet

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